



Ollscoil  
Teicneolaíochta  
an Atlantaigh

Atlantic  
Technological  
University

B.Sc. (Hons) in Software Development  
**Applied Project & Minor Dissertation**

## Project Proposal

Each student must complete the following form and submit it to their supervisor for consideration. Once your supervisor has signed-off on the proposal, you must submit the document for consideration using Moodle.

1. **Student Name:** Ryan Curley
2. **Supervisor Name:** Patrick Beatty
3. **Project Name:** Media Store

4. **Project Context**

Describe the context of the problem domain here. Explain what you are proposing to do and your rationale for doing it. Explain why the problem domain is of interest.

The context of the problem domain for my project is to create a web app to simulate payment processes using my own games and music I have created over the years. I am proposing to create a real-world retail application where a customer, employee and manager will login and different items will be displayed to them. A customer will see available games and music, be able to add games to a wish list and view their purchase history which will then use AI to provide personal recommendations based on recently bought games. An employee will be able to view orders which will involve tracking customer transactions like games and/or music bought. A manager will be able to view available stock, sold products and customer summary. The problem domain is of interest because I want to make an application that will involve as much as myself into it as possible, so I can create something I care about.



Ollscoil  
Teicneolaíochta  
an Atlantaigh

Atlantic  
Technological  
University

## 5. Project Objectives

Write out the key objectives of the project as bullet points. Each objective should be clear, realisable and measurable / testable, i.e. the success of your project is determined by the degree to which these are realised.

- To create simulated payments that demonstrates real world retail transactions.
- To utilise various cloud technologies to make the application as scalable and secure as possible.
- To use automated testing to ensure speed, reliability and early bug detection.
- To effectively use test driven development and CI/CD to verify all possible bugs are prevented.
- To use external APIs to create personalised recommendations for the customer so to utilise AI technologies.

## 6. Technologies & System Architecture

Explain the technologies you are going to use and why you selected them. These include the programming languages, operating systems, presentation and storage technologies and any cloud / 3rd party libraries / services that you intend to use.

JavaScript with React for code reusability and immediate browser updates.  
Java with Spring Boot for multithreaded tasks for application efficiency.  
AWS EC2 for load balancing to prevent sever spikes.  
AWS S3 for secure storage and file delivery.  
Azure for CI/CD to ensure code is tested and built during every commit.  
Database for AWS RDS for the storing of structured data like orders and payments.  
Postman - Testing API calls in the backend.  
External APIs – FastAPI for personalised recommendations, PayPal Sandbox API for simulated payments.  
Testing – JUnit for unit and integration tests, JaCoCo for code coverage verification and Postman for API testing.



## 7. Schedule of Work

Using a Gantt chart or tabular format, outline your schedule of work for all the key project activities, deliverables and dates.

| Task No | Phase       | Task / Deliverable          | Description                                | Start Date | End Date | Duration | Dependencies |
|---------|-------------|-----------------------------|--------------------------------------------|------------|----------|----------|--------------|
| 1       | Development | Backend and Frontend Setup  | Create initial structure, connect to DB    | Oct 17     | Oct 17   | 1 day    | —            |
| 2       | Development | Create Microservices        | Accounts, Orders, Payments, Products, etc. | Oct 18     | Oct 19   | 1 day    | 1            |
| 3       | Development | Implement API Endpoints     | CRUD operations and authentication         | Oct 20     | Oct 22   | 2 days   | 2            |
| 4       | Testing     | Unit & Integration Tests    | JUnit, JaCoCo, and Postman API testing     | Oct 22     | Oct 29   | 7 days   | 3            |
| 5       | Development | Integrate PayPal Sandbox    | Simulated payments                         | Oct 29     | Nov 5    | 7 days   | 4            |
| 6       | Development | AI Recommendation Service   | Integrate model and API                    | Nov 5      | Nov 12   | 7 days   | 5            |
| 7       | Development | Connect Frontend to Backend | Fetch API → Spring Web                     | Nov 12     | Nov 14   | 2 days   | 6            |
| 8       | Development | Frontend UI Development     | Build React components and pages           | Nov 14     | Nov 19   | 5 days   | 7            |
| 9       | Testing     | Frontend Testing            | UI/UX testing with React Testing Library   | Nov 19     | Nov 26   | 7 days   | 8            |
| 10      | Deployment  | CI/CD Setup (Azure)         | Automate builds, testing, deployment       | Nov 26     | Dec 01   | 5 days   | 9            |